Deposit Insurance

Lecture 10

BANK3011
DR WEI CUI

THE UNIVERSITY OF SYDNEY
Business School



Objectives of this week's lecture

- Liquid asset and liability management
- Objectives of Deposit Insurance and Govt Guarantees
- Controlling Risk Taking
- Australia's recent experience during the Global Financial Crisis (see RBA articles on BB):
  - ✓ Financial Claims Scheme
  - ✓ Australian Government Wholesale Funding Guarantee Scheme
- Fed's Discount Window

0. Liquid asset and liability management

- The Difficulties in Managing Liquidity
  - ✓ The problem is that the level of liquidity in assets and liabilities is inversely correlated to return
    - ➢ More liquid assets have lower returns, whilst longer terms assets have higher returns
    - ➢ More callable (subject to withdrawal) liabilities have lower costs, whilst more stable liabilities have higher costs

0. Liquid asset and liability management

- Liquid asset management
  - ✓ Liquid assets are traded in a deep market
    - ➢ Even large trading volumes will not impact on price
    - ➢ Liquid assets include Treasury notes and Treasury bonds
  - ✓ Liquid securities carry a comparatively lower interest rate: risk–return trade-off

0. Liquid asset and liability management

- The composition of an FI's liquid asset portfolio, especially with respect to cash and government securities, is partly determined by:
  - ✓ earnings consideration
  - ✓ minimum liquid-asset reserve requirements imposed by the central bank
- Historically, many countries had imposed reserve ratios
- Government securities act as secondary reserves (buffer reserves)

0. Liquid asset and liability management

- Managing the securities portfolio is an integral part of liquidity management for financial institutions
- The conflict between and the balance of the risk of illiquidity and the desire to maintain a high investment return is at the heart of management of the liquid asset portfolio
- The determinants of an FI's optimal liquid asset ratio depend on a number of factors:
  - The variability of deposit inflows and outflows
  - The yield on non-liquid assets.
  - The liquidation costs of highly non-liquid assets.
  - Higher interest rates paid on purchased funds/liabilities in the short-term money market, or in utilising the RBA liquidity facilities, will increase the desired liquidity asset/cash ratio

0. Liquid asset and liability management

- The cash market and exchange settlement account (ESA)
  - The cash market is where banks lend and borrow funds from each other, typically on an overnight basis. These funds can be used to:
  - Meet liquidity needs: Banks may borrow funds to ensure they meet short-term liquidity requirements, such as the ESA balances.
  - Support lending activities: Banks may also use borrowed funds to support ongoing lending activities.
  - Facilitate settlement of transactions: The funds can help banks settle large-scale interbank transactions efficiently.
- The price of funds in this market is the interest rate on these interbank loans, which is called the **cash rate** in Australia.
- The quantity traded in this market is called the **Exchange Settlement (ES) balance** - the balance of DI's deposit accounts at the RBA, the ESA.

0. Liquid asset and liability management

- RBA sets a target for the cash rate, it is often referred to as a “tool” of monetary policy.
  - The RBA managed the supply of ES balances so that it met estimated demand and the cash rate was close to its target.
  - The RBA achieved cash rate target through open market operations, particularly through repurchase agreements (repos).

Example: On Monday, the RBA conducts a 7-day repo with Bank A. The RBA buys government bonds from Bank A for $80,000,000 and credits Bank A’s Exchange Settlement (ES) account for the full purchase amount. The repo rate announced by the RBA for this operation is 3.00% p.a. (actual/365). The repo matures the following Monday. Calculate the repurchase price Bank A must pay to unwind the repo on maturity.

0. Liquid asset and liability management

Example: On Monday, the RBA conducts a 7-day repo with Bank A. The RBA buys government bonds from Bank A for $80,000,000 and credits Bank A's Exchange Settlement (ES) account for the full purchase amount. The repo rate announced by the RBA for this operation is 3.00% p.a. (actual/365). The repo matures the following Monday. Calculate the repurchase price Bank A must pay to unwind the repo on maturity.

$$\text{Repurchase Price} = \text{Purchase Price} \times \left(1 + \text{repo rate} \times \frac{\text{days}}{365}\right)$$

$$= \$80{,}000{,}000 \times \left(1 + 0.03 \times \frac{7}{365}\right) = \$80{,}000{,}000 \times 1.00057534246575 \approx \boxed{\$80{,}046{,}027.40}$$

- The RBA's buy-side repo credits Bank A's ES balance by $80m, increasing system ES supply. With more ES liquidity, banks have less need to borrow overnight, so the interbank cash rate falls.

0. Liquid asset and liability management

- Liability liquidity management
  - ✓ Overall aim of liability management: to construct a low-cost and low–withdrawal-risk liability portfolio
    - ➢ More callable (subject to withdrawal) liabilities have lower costs, whilst more stable liabilities have higher costs

0. Liquid asset and liability management

- Deposit liabilities
  - Cheque account and other demand deposits
  - Savings account
  - Fixed-term deposits
  - Negotiable certificates of deposits (NCDs)

- Non-deposit liabilities
  - Interbank funds
  - Repurchase agreements (Repos)
  - Bank accepted bills
  - Commercial bills or non-bank bills
  - Commercial paper or promissory notes
  - Subordinated debt, medium-term notes and long-term borrowings

…

0. Liquid asset and liability management

- The withdrawal risk and funding cost characteristics of the major liabilities of a modern DI
  - ✓ Demand deposits
    - ➢ High degree of withdrawal risk
    - ➢ Low costs
    - ➢ In U.S., demand deposits paid zero explicit interest!
    - ➢ This does not mean that demand deposits are free to DIs.

Consider the following case:

- A depositor holds on average $250 per month for the first three months of the year, $500 per month for the next three months, and $1000 per month for the final six months of the year in a demand deposit account.
- The account pays 5 per cent per annum if the minimum balance is $500 or more, and it pays no interest if the account falls below $500.
- The depositor makes an average of 50 internet transactions per month and pays a service fee of 10 cents for each transaction, although it costs the bank 15 cents to process each transaction.

The account holder's gross interest return, consisting of implicit plus explicit interest, is:

Gross interest return = explicit interest + implicit interest

=$500(0.05)(0.25)+$1000(0.05)(0.5)+($0.15−$0.10)(50)(12) =$6.25+$25+$30=$61.25

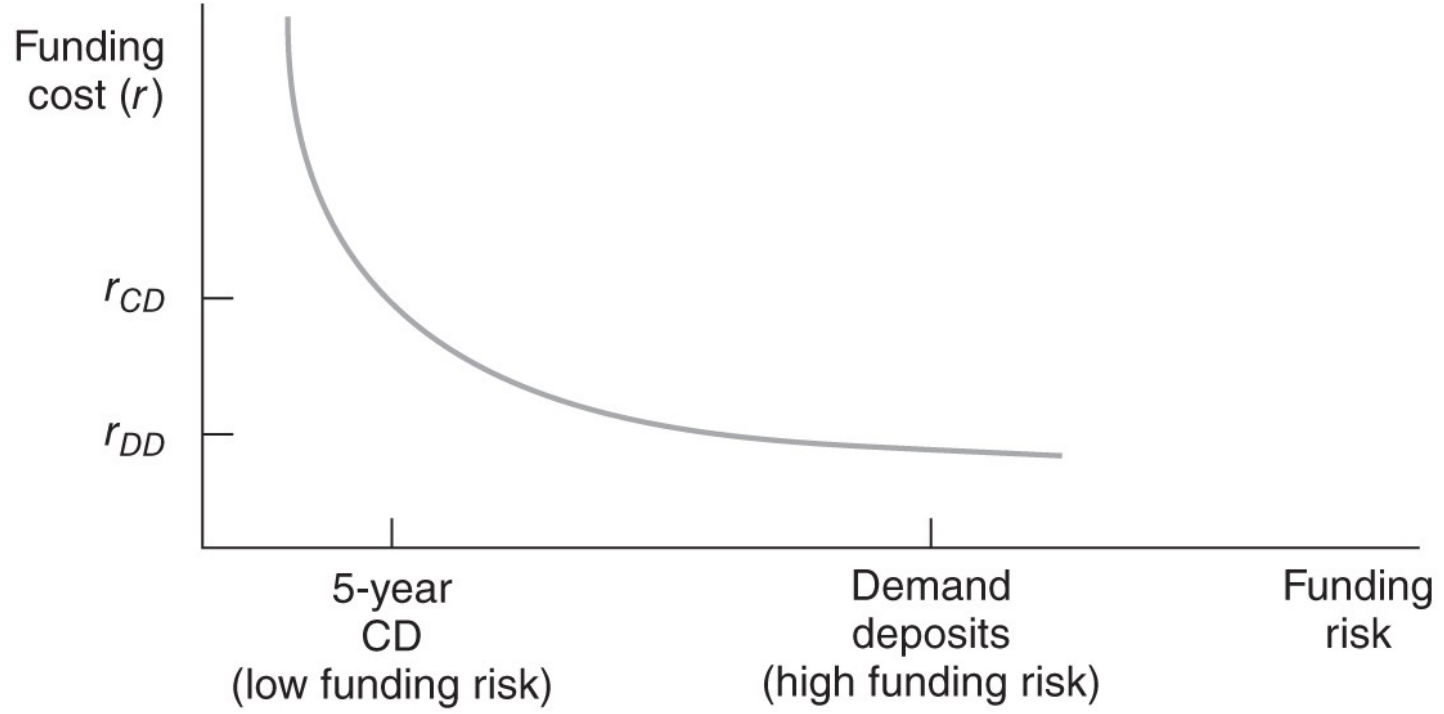
0. Liquid asset and liability management

- The withdrawal risk and funding cost characteristics of the major liabilities of a modern DI
  - ✓ Interbank funds
    - Short-term un-collateralised loans between FIs
    - Rollover risk (Market stress, loss of confidence, counterparty concers → lenders pull back.)
    - Cost: short-term money market rate
  - ✓ Repos
    - Collateralised interbank transactions
    - Highly liquid and flexible source of funds
    - Yield generally below the market interbank rate due to collateralised nature

0. Liquid asset and liability management

- Funding Risk and Cost
  - ✓ A DI must trade off the benefits of attracting liabilities with a low funding cost with a high chance of withdrawal against liabilities at a high funding cost and low liquidity.

Copyright © 2014 McGraw-Hill Education. All rights reserved. No reproduction or distribution without the prior written consent of McGraw-Hill Education.



0. Liquid asset and liability management

- In Sum, Liquid Asset and Liability Management
  - ✓ A DI manager can optimize over both liquid assets and liability structures to insulate the DI against liquidity risk.
    - ➢ On the one hand, the DI manager needs to build up a prudential level of liquid assets while minimizing the opportunity costs of funds. A DI manager should try to achieve an optimal mix of lower-yielding, liquid assets and higher-yielding, less liquid assets. Holding too many liquid assets penalizes earnings, while holding too few liquid assets exposes the FI to enhanced liquidity crises.

0. Liquid asset and liability management

- In Sum, Liquid Asset and Liability Management
  - ✓ A DI manager can optimize over both liquid assets and liability structures to insulate the DI against liquidity risk.
    - ➢ On the other hand, the DI manager needs to structure the liabilities so that the need for large amounts of liquid assets is reduced while trading off funding risk and funding costs.

1. Deposit insurance

- Remember this?



1. Deposit insurance

- Demand deposit contracts are first-come, first-served contracts. A depositor either gets paid in full or nothing. When a DI's assets are valued at less than its deposits, only certain proportion of the depositors will be paid in full and a depositor's place in line determines the amount he or she will be able to withdraw from the DI. Knowing this, any line outside a DI encourages other depositors to join the line immediately even if they do not need cash today for normal consumption purposes.

1. Deposit insurance

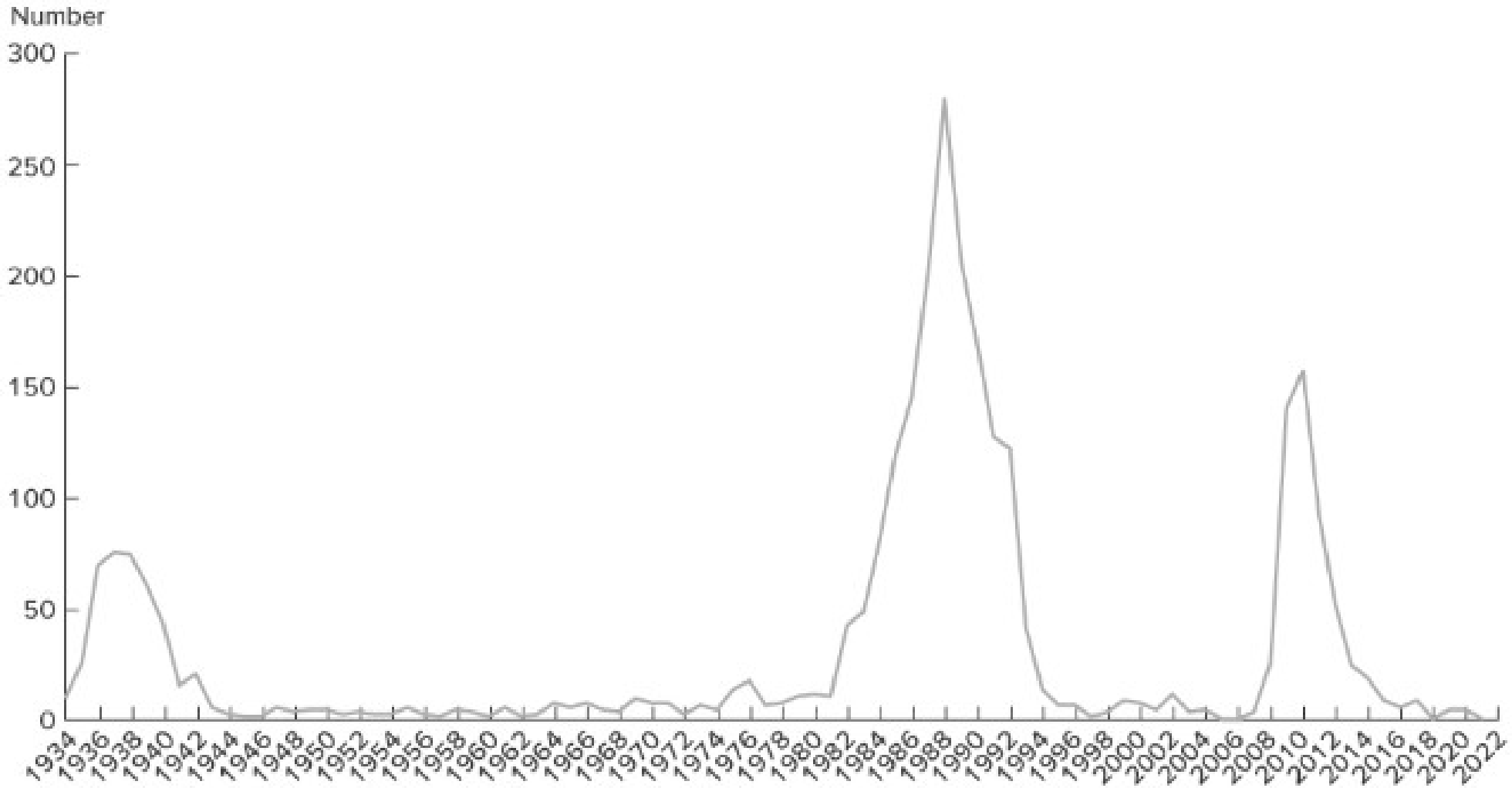
- The incentives for depositors to run first and ask questions later creates a fundamental instability in the banking system in that an otherwise sound DI can be pushed into insolvency and failure by unexpected large depositor drains and liquidity demands.

- Regulators have recognized this inherent instability of the banking system and put in place mechanisms to ease the problem.

1. Deposit insurance

- USA:
  - ✓ Federal Deposit Insurance Corporation (FDIC) created in 1933
  - ✓ Federal Savings and Loan Insurance Corporation (FSLIC)
- FDIC deposit insurance covers the depositors of a failed FDIC-insured depository institution dollar-for-dollar, principal plus any interest accrued or due to the depositor, through the date of default, up to at least $250,000 per depositor, per FDIC-insured bank, per ownership category.
- If a deposit holder believes a claim is totally secure, even if the DI is in trouble, the holder has no incentive to run.

1. Deposit insurance



Source: FDIC annual reports and statistics on banking. *www.fdic.gov*

1. Deposit insurance

- FDIC Insolvency
  - ✓ Two schools of thought:
  - ✓ FDIC insolvent or near-insolvent a number of times in its history, due to:
    - ➢ Financial environment and prevailing economic conditions
      - ▪ Rise in interest rates, Collapse in oil, real estate, and commodity prices
    - ➢ Late 2000s:
      - ▪ Housing market collapse and demise of high profile FIs
      - ▪ Consumer loan defaults

1. Deposit insurance

- FDIC Insolvency
  - ✓ Two schools of thought:
  - ✓ FDIC insolvent or near-insolvent a number of times in its history, due to:
    - ➢ And moral hazard
      - ▪ Deposit insurance reduces market discipline
      - ▪ Premiums not linked to risk

1. Deposit insurance

- Causes of depository fund insolvencies
  - ✓ Moral hazard
    - ➢ Issue:
      - ▪ Premiums not linked to risk
        - • Insurance is not actuarially fairly priced
      - ▪ Inadequate monitoring
        - • No incentive for depositor discipline (eg. Through fund withdrawal, required deposit rates)
    - ➢ Result: Encourage rather than discourage risk taking
      - ▪ Incentive to take excessively risky investment decisions
  - ✓ The tradeoff between bank run risk vs moral hazard risk

2. Controlling DI Risk Taking

- Methods For Controlling DI Risk Taking

✓ The moral hazard problems of deposit insurance can be mitigated by:

➢ 1. Increasing stock holder discipline

➢ 2. Increasing depositor discipline

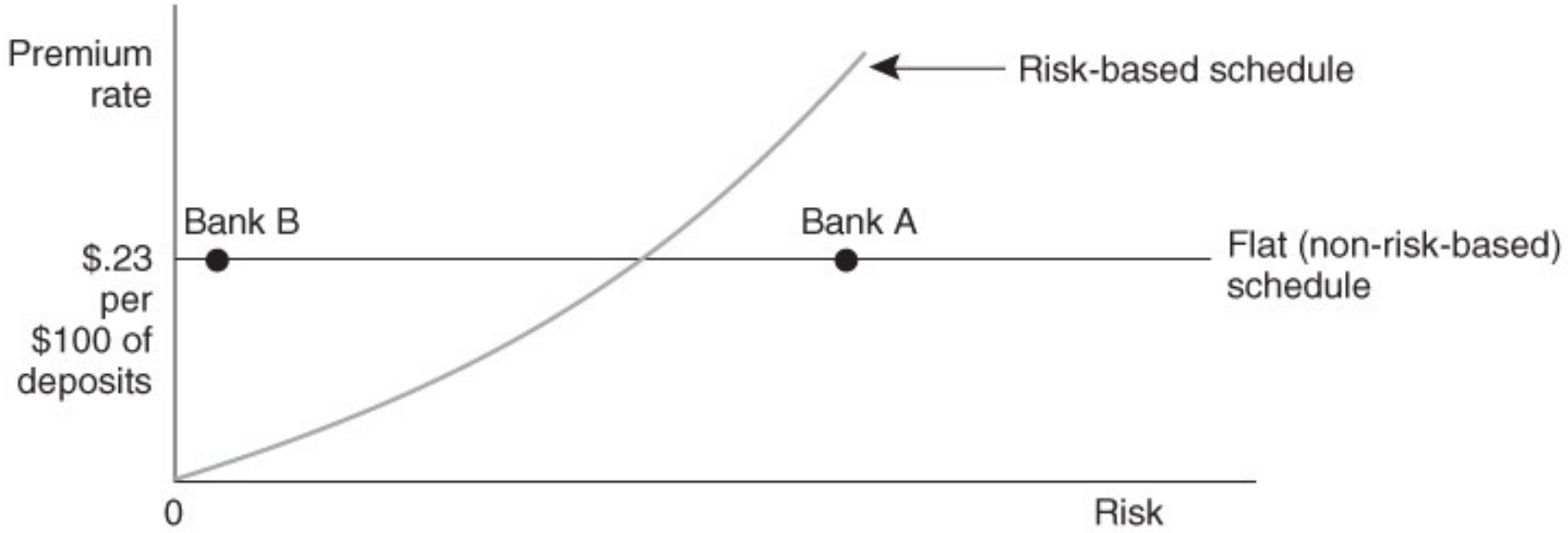
➢ 3. Increasing regulator discipline

2. Controlling DI Risk Taking

- Controlling FI Risk Taking

  1. Stockholder discipline

     - Risk not always controlled by the shareholders or factored in by the regulator:



     - Link insurance premiums to risk

2. Controlling DI Risk Taking

- Risk-Based Deposit Insurance
  - ✓ Implement risk (factor)-based premiums based on:
    - ➢ Categories and concentrations of assets
    - ➢ Categories and concentrations of liabilities
    - ➢ Other factors that affect probability of loss
    - ➢ Deposit insurer’s revenue needs

2. Controlling DI Risk Taking

Table Summary: A table with 4 columns and 2 sections. Column 1 titled 1 capital category lists the categories, while columns 2 to 4 titled 2 healthy, 3 supervisory concern, and substantial supervisory concern list the corresponding values. The top section is titled Plan A and the bottom section is titled Plan B.

Supervisory Groups—Premiums

(1) Capital Category	(2) Healthy[1]	(3) Supervisory Concern[2]	(4) Substantial Supervisory Concern[3]
Panel A: The Fee Structure for Deposit Insurance, Effective January 1, 1993			
Well capitalized[4]	23 cents per $100	26 cents per $100	29 cents per $100
Adequately capitalized[5]	26 cents per $100	29 cents per $100	30 cents per $100
Undercapitalized[6]	29 cents per $100	30 cents per $100	31 cents per $100
Panel B: The Fee Structure for Deposit Insurance, Effective January 1, 1997			
Well capitalized[4]	0 cents per $100	3 cents per $100	17 cents per $100
Adequately capitalized[5]	3 cents per $100	10 cents per $100	24 cents per $100
Undercapitalized[6]	10 cents per $100	24 cents per $100	27 cents per $100

2. Controlling DI Risk Taking

- Increased capital requirements, stricter DI closure rules
  - ✓ Higher capital requirements
    - ➢ So stockholders have more at stake in taking risky investments
    - ➢ Five capital zones with progressively harsher mandatory actions being taken by regulators as capital ratios fall (e.g. a bank is placed into receivership within 90 days of the time when its capital falls below some book value level – currently 2% of assets).
  - ✓ Stricter closure rules
    - ➢ Controls forbearance

2. Depositor Discipline

- ✓ Aim: create conditions for a greater degree of depositor discipline, eg. Depositors require higher interest rates to deposit funds, or ration the amount of deposits they are willing to hold in riskier Dis

- ✓ Insurance cap: above the cap amount, no insurance is provided
  - ➢ Can be bypassed by altering the structure of deposit funds and by spreading deposits across funds
  - ➢ Too-big-to-fail: implicit coverage without cap

2. Depositor Discipline

✓ Insured depositor transfer:

- Insured deposits are transferred in full to another DI
- Uninsured depositors take a loss (haircut) on DIs failure

3. Regulatory Discipline:

- ✓ Frequency and thoroughness of Examinations:
  - ➢ Improved accounting standards including market valuation of assets and liabilities
  - ➢ Annual on-site examination of every bank
  - ➢ Independent audits

- ✓ Capital forbearance:
  - ➢ Prompt corrective action

3. Others

- Another mechanism: Discount Window
  - ✓ Discount window = central bank lender of last resort facility
    - ➢ Short-term, non-permanent
    - ➢ Requires high-quality liquid assets as collateral
    - ➢ To borrow from the discount window, a DI generally needs high-quality liquid assets to pledge as collateral. The interest rate charged on the loans is called the discount rate.

3. Others

- Another mechanism: Discount Window
  - ✓ Lending programs are now offered by the Federal Reserve Bank:
  - ✓ Primary credit: available to sound DIs
    - ➢ Very short term (typically overnight) at a rate above the Federal Funds rate.
    - ➢ May be used for any purposes
    - ➢ Can be extended for periods of up to few weeks

3. Others

- Another mechanism: Discount Window
  - ✓ Secondary credit: for DIs not qualified for primary credit
    - ➢ Borrowing rates are above primary credit rate
    - ➢ Temporary source to meet liquidity needs. Its use should be consistent with a timely return to a reliance on market sources of funding or the orderly resolution of a troubled institution.
    - ➢ May not be used to fund an expansion of the borrower's assets

4. The Australian Experience

- Prior to 12 October 2008, Australia was one of the only two OECD countries (New Zealand being the other) without an explicit deposit insurance scheme.
  - ✓ Emphasis on bank regulation by APRA to protect deposit holders (with depositor preference)
  - ✓ Deposit insurance viewed as expensive and incentive-perverting

- Explicit deposit guarantee arrangements were introduced by the Australian Federal government in October 2008 in response to the extremely difficult conditions in the global financial system during the GFC.

4. The Australian Experience

- Australia had no choice but to provide similar support for our banks to help them remain competitive internationally as other countries strengthen their explicit deposit insurance schemes and/or provided government guarantees on bank debt.

4. The Australian Experience

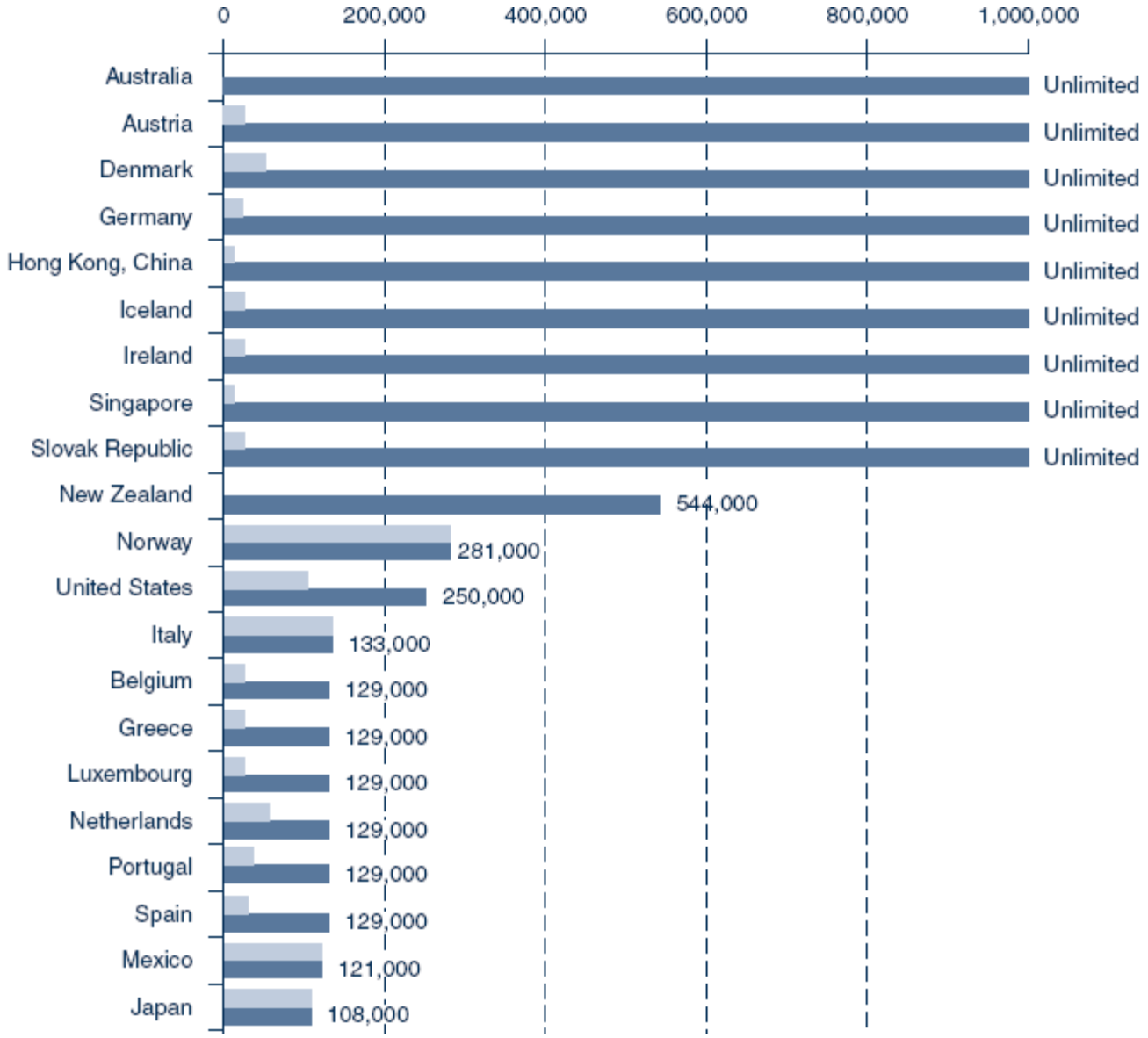
- There are two schemes that were introduced under the Australian arrangements;

  ✓ Under the Financial Claims Scheme (FCS), all deposits up to AUD 1 million were automatically guaranteed at NO COST by the government, from February 2012 this was reduced to AUD 250,000

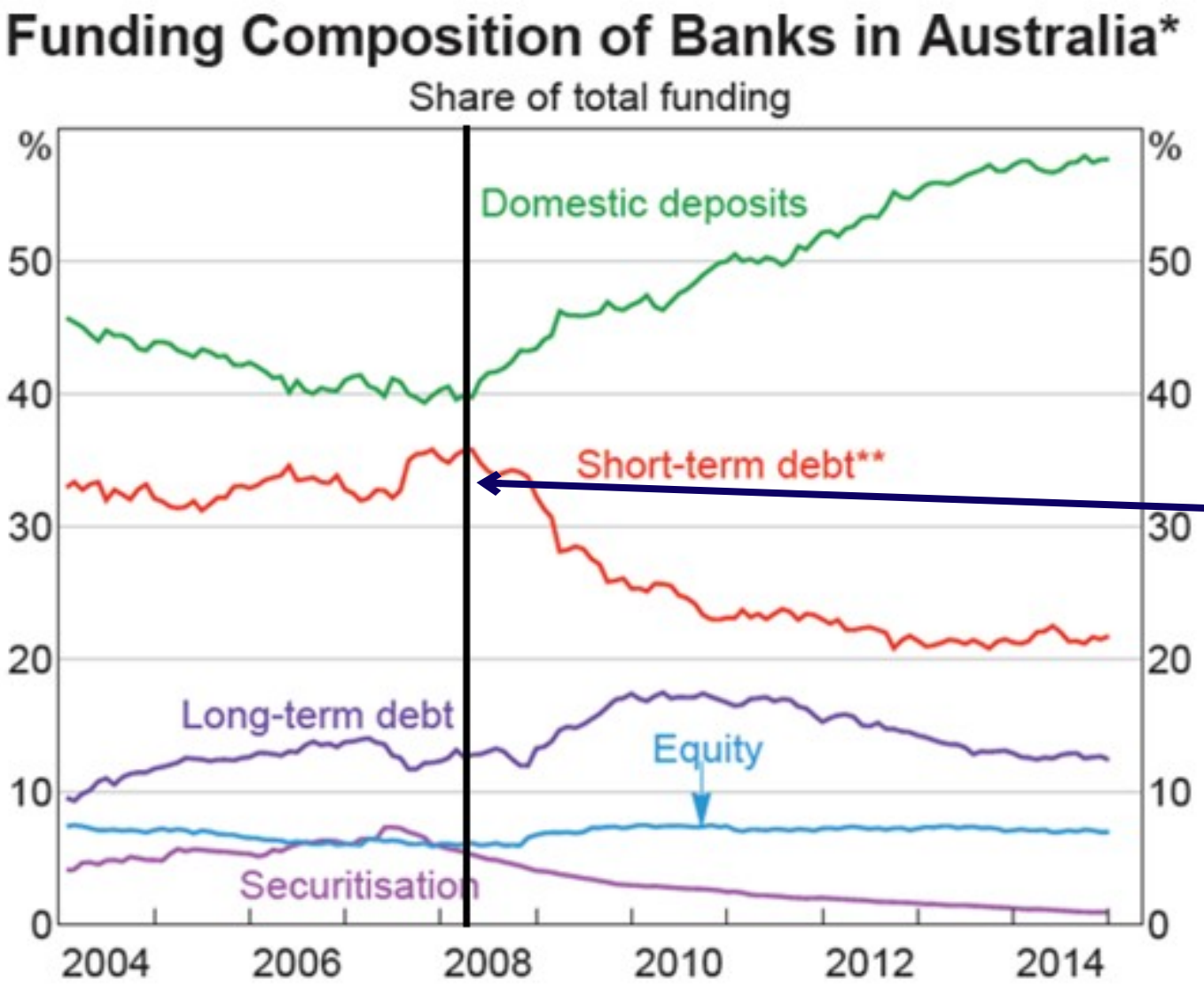
  ✓ The Australian Government Guarantee Scheme for Large Deposits and Wholesale Funding (WGS), where eligible ADIs could pay a risk-based FEE for a government guarantee on deposits greater than AUD 1 million and on wholesale debt funding.

4. The Australian Experience

- Deposit insurance before and after 2008



4. The Australian Experience



Introduction of the Explicit Deposit a
Wholesale Funding Guarantee

Source: APRA